

Theme: Social Media Profile Manager

Section A: Conceptual Understanding

1. Explain the Django Request-Response cycle and how it differs from a standard Python script execution.

- In Django, a user sends a request through a browser.
- Django processes the request using URLs, Views, Models, and Templates.
- Finally, a response is returned to the user.
- A standard Python script runs from start to end without handling web requests.

2. Explain why Django Model Fields (CharField, IntegerField) are more robust for profile data than Python dynamic typing.

- Django Model Fields enforce data types and validation.
- CharField stores text, while IntegerField stores numbers only.
- This reduces data errors and improves database consistency.
- Python variables can change types dynamically, which may cause unexpected issues.

3. Explain how Django Forms handle automated input validation for usernames and age ranges.

- Django Forms automatically validate user input.
- They check required fields, data types, and field lengths.
- Custom validation methods can enforce age limits or username rules.
- Invalid data generates error messages automatically.

4. Explain how to implement conditional logic in Django Templates to toggle account visibility.

- Django Templates use `{% if %}` statements for conditions.
- Example:

```
{% if profile.is_public %}
Public Profile
{% else %}
Private Profile
{% endif %}
```
- This displays content based on the profile visibility status.

5. Explain the difference between iterating through a Python list and a Django QuerySet.

- A Python list stores data in memory.
- A Django QuerySet retrieves data from the database.
- QuerySets are lazy-loaded, meaning data is fetched only when needed.
- Both can be iterated using loops, but QuerySets interact with the database.

6. Explain why the Django ORM is preferred over Python dictionaries for persistent profile storage.

- Django ORM stores data permanently in a database.
- It provides CRUD operations (Create, Read, Update, Delete).
- Python dictionaries store data temporarily in memory.
- ORM offers security, scalability, and easier database management.